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LETTER

Variable phenology but consistent loss of ice cover on 1213 Minnesota lakes

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Scientific Significance Statement

Lake ice cover is declining globally due to climate change. Linear trend analysis of a small number of well-studied lakes has been the primary method for quantifying rates of ice loss, often ignoring the potential for increasing variance, accelerating trends, and sensitivity to the time frame of analysis. Using 74 years of data for 1213 lakes in Minnesota, USA, we confirm that regional ice loss has likely been linear (−2 d per decade) with increasing variance over time that could be driving recent reports of accelerating ice loss estimated by comparing shorter time intervals within individual lakes.

Abstract

Lake ice cover is declining globally with important implications for lake ecosystems. Ice loss studies often rely on small numbers of lakes with long-term data. We analyzed variation and trends in ice cover phenology from 1213 lakes over 74 yr (1949–2022) in Minnesota (USA), during which ice cover duration declined at a rate of 2 d per decade (14 d total) and became more variable. Despite variation in phenology, just 10–20% of lakes differed from statewide phenological trends. Accounting for synchronous annual variation and estimating trends over long time periods (e.g., > 40 yr) were critical for obtaining robust

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estimates of ice loss. The constant rates estimated here were consistent with recent global estimates (1.7–1.9 d per decade) and suggest that, even if present, accelerating rates of ice loss could be difficult to detect in the midst of shorter-term periods of warming and increasing variability.

Climate change affects lake thermal regimes in numerous ways, and responses vary across lakes and seasons (e.g., O'Reilly et al. 2015; Winslow et al. 2015). For example, ice cover influences lake abiotic and biotic processes (Hampton et al. 2017), from lake temperatures in subsequent seasons (O'Reilly et al. 2015; Dugan 2021) to the reproductive success of sport fish (Farmer et al. 2015; Lyons et al. 2015; Lynch et al. 2016). Declining and more variable ice cover due to climate change can thus have far-reaching implications for lake ecological processes in both the short and long term (Adrian et al. 1999; Magnuson et al. 2000; Hampton et al. 2017; Feiner et al. 2022).

Ice cover duration is dictated by the timing of both ice formation and ice breakup. Each event is influenced by different climatic processes interacting with lake characteristics, and each has distinct implications for lake processes. Ice formation is controlled by local winds and air temperature and depends on lake area, depth, and fetch (Woolway et al. 2020). Ice formation influences under-ice temperature, oxygen dynamics, and over-ice snow cover (Kirillin et al. 2012), thereby influencing dissolved oxygen, under-ice light conditions, primary production, and the likelihood of fish kills (Greenbank 1945; Hampton et al. 2017). Ice breakup is influenced by air temperature, wind, precipitation, snow cover, and solar radiation (Kirillin et al. 2012; Preston et al. 2016). The date of ice breakup is a key driver of spring and summer water temperature and oxygen dynamics (Magee et al. 2016, 2019), zooplankton production (Preston and Rusak 2010), and spawning timing and success of certain species of fish (Schneider et al. 2010; Lyons et al. 2015; Lynch et al. 2016).

Declines in the duration of ice cover and changing ice phenology have been documented for lakes in the Northern Hemisphere. Ice duration is shortening, ice formation is happening later, and ice breakup is happening earlier, although variable trends and magnitudes of change have been reported across different geographical locations, time periods, and methods of analysis (Brown and Duguay 2010). For example, two recent global analyses of long-term (1855–2019; 1815–2018) trends showed ice duration declined 1.9 d per decade (19 lakes in Woolway et al. 2020 and 18 lakes in Imrit and Sharma 2021, with 10 shared lakes), in contrast to shorter-term studies documenting ice duration declining over twice as quickly in the late 20th century (e.g., 4.3–5.3 d per decade; Jensen et al. 2007; Benson et al. 2012). Also, recent studies (e.g., Woolway et al. 2020) report faster rates of ice loss than past studies (e.g., Magnuson et al. 2000), and studies with more recent observations have reported accelerating ice cover loss in recent time periods (e.g., Sharma et al. 2021).

Rates of ice cover loss have been estimated primarily through linear trend analysis, often ignoring potential changes in variance or rates of change over time (Jensen et al. 2007; Benson et al. 2012; Hewitt et al. 2018), with some exceptions for accelerating rates over time (Sharma et al. 2021; Imrit and Sharma 2021). Accounting for these changes could provide a more complete understanding of lake ice cover change and more robust estimates of ice cover loss. For example, differences in estimated rates of ice loss could be due to nonlinear and possibly accelerating rates of ice loss in recent decades (Weyhenmeyer et al. 2004; Johnson and Stefan 2006; Magee et al. 2016). Alternatively, interannual variability in ice cover is increasing (Feiner et al. 2022), which could influence overall trend estimates and the perception of accelerating change, depending on the time frame of analysis (Magnuson 1990; Benson et al. 2012).

Here, we analyzed long-term trends and variability in ice cover duration, ice formation date, and ice breakup date using data from 1213 lakes in Minnesota, USA, collected by community scientists and government agencies from 1949 to 2022. This dataset allowed us to explore spatiotemporal ice cover dynamics in a lake-rich region, complementing global studies using a small number of well-documented cases. Our objectives were to quantify (1) potentially nonlinear statewide trends in each metric, (2) lake-specific trends, and (3) the sensitivity of results to the time frame of analysis.

Methods

Data collation

Ice formation and breakup observations were recorded by volunteers throughout Minnesota from 1949 to 2022 and collated and maintained by the Minnesota Department of Natural Resources State Climatology Office. In some cases, multiple dates of ice formation or ice breakup were recorded for a given year, and we used the latest date of ice formation and the earliest date of ice breakup to calculate a conservative estimate of ice cover duration. There is no standard definition for ice formation and breakup phenology, though definitions are often consistent within lakes (Sharma 2022), as is the case with these data. This approach yielded a total of 8778 lake-years with paired formation and breakup data from 1213 lakes which formed the primary dataset for analysis (Walsh et al. 2024). Because some lakes in our dataset were only observed once, we assessed the sensitivity of our results to the duration and number of observations within individual lakes by repeating analyses for lakes with at least 10 yr of paired data (262 lakes; 5601 lake-years) and lakes with at least 30 yr of paired data (54 lakes; 2247 lake-years). Similarly, because ice breakup was more frequently recorded than ice formation

(1810 lakes; 18,636 lake-years), we repeated the ice breakup analysis for the larger dataset for comparison.

Trend analysis overview

We used generalized additive mixed models (Simpson 2018; Pedersen et al. 2019) to quantify potentially nonlinear trends in ice cover duration and the timing of ice formation and breakup. We used the packages "mgcv" (Wood 2015) and "gratia" (Simpson and Singmann 2023) in R version 4.3.1 (R Core Team 2023) to evaluate trends and significant periods of change. We fit the same model form for each of the three ice metrics (duration, formation, and breakup), where we assumed a given ice metric for lake i in winter year t was normally distributed with expected value, $\mu_{i,t}$, and standard deviation, $\sigma_{i,t}$ (estimated separately for each metric).

$$\text{Ice metric}_{i,t} \sim N\left(\mu_{i,t}, \sigma^2_{i,t}\right)$$

$$E\left(\text{Ice metric}_{i,t}\right) = \mu_{i,t}$$

$$\text{var}(\text{Ice metric})_{i,t} = \sigma^2_{i,t}$$

$$\begin{aligned}\mu_{i,t} = {} & f\left(\text{winter year}_t\right) + f\left(\ln \text{lake area}_i\right) + f\left(\text{UTM}\,X_i, \text{UTM}\,Y_i\right) \\ & + f\left(\text{ENSO}_t\right) + f\left(\text{QBO}_t\right) + f\left(\text{PDO}_t\right) + f\left(\text{NAO}_t\right) \\ & + f\left(\text{solar cycles}_t\right) + \text{Year}_t + \text{Lake}_i\end{aligned}$$

$$\text{Year}_t = N\left(0, \sigma^2_{\text{Year}}\right)$$

$$\text{Lake}_i = N\left(0, \sigma^2_{\text{Lake}}\right)$$

$$\log E\left(\sigma_{i,t} + 0.01\right) = f\left(\text{winter year}_t\right) + f\left(\text{UTM}\,X_i, \text{UTM}\,Y_i\right)$$

The mean value for lake i in winter year t ($\mu_{i,t}$) is the sum of smoothed functions of winter year, log-transformed lake area (Minnesota Department of Natural Resources 2021), the UTM Zone 15 coordinates of each lake centroid, and multiple climate indices (El Niño-Southern Oscillations [ENSO], Quasi-Biennial Oscillation [QBO], Pacific Decadal Oscillations [PDO], North Atlantic Oscillations [NAO], solar cycles; refer to the Supporting Information for details). The models also include random intercept effects for winter year as an unordered factor (Year_t) and lake ID (Lake_i). Including random intercepts for winter year allows the model to separate random annual variation from the long-term trend (sensu Knape 2016). We fit a post hoc model of the estimated annual random intercepts as a smoothed function of continuous winter year to test for remaining temporal patterns in the data. To allow for potential spatial and temporal trends in variance (e.g., due to increasing variance over time or spatial variation in the sampling scheme), we fit models with and without a heteroskedastic term wherein standard deviation was modeled as a smoothed function of winter year and lake centroid coordinates. All smooth terms used thin plate regression spline smoothing basis functions and were "tuned" to set the "wiggliness" (using k, the upper limit on the number of basis functions included in the model) to the minimum value that would pass a "gam.check()" test (Wood 2015). Shrinkage penalties were added to each term so that predictors could be penalized out of the model if they did not improve fit (Marra and Wood 2011). Models were fit using maximum likelihood, using Akaike information criterion to compare models with and without the spatial and temporal heteroskedasticity. Restricted maximum likelihood was used for final model fits, predictions, figures, and analyses. Terms that were insignificant or penalized out of the maximum likelihood models were not included in the final models.

Statewide trends

We estimated statewide trends, trend derivatives, and associated "simultaneous" confidence intervals (CIs; to represent uncertainty around the location of the curve rather than at points along the curve) using the functions "confint()" and "derivatives()" in the package "gratia" (Simpson and Singmann 2023). To summarize change over the entire time period, we computed average estimates from the final statewide models for the five earliest years (1949–1953) and five most recent years (2018–2022) with all covariates except winter year excluded from prediction. We estimated the difference in these 5-yr averages (69 yr between the midpoints of each period) and computed 95% CIs using posterior simulation from the final models. We used the estimated degrees of freedom for the winter year smooth term to quantify nonlinearity in the long-term trend.

To investigate trends in variability in each metric, we (1) calculated the 5-yr rolling standard deviation across the raw metrics for all lakes, (2) calculated the 5-yr rolling standard deviation of the modeled winter year random intercepts, (3) plotted the modeled smooth function of the standard deviation over winter year, and (4) calculated the 5-yr rolling standard deviation of model residuals. We examined spatial patterns in model predictions in the year 2022 for each ice metric using semivariograms using the "Variogram()" function in the package "nlme" (Pinheiro et al. 2023).

Lake-specific trends

To test whether lake-specific trends differed from the statewide trend, we fit a hierarchical GAM by adding a factor-smooth interaction between winter year and lake, where lake-specific winter year smoothers were allowed to have differing basis dimension complexity (i.e., … + $f(\text{winter year}_t, \text{lake}_i)$; Pedersen et al. 2019). We used a smaller subset of higher quality time series from 1970 to 2022 to fit these models, including all lakes since 1970 with at least 10 lake-years of data (these years did not need to be consecutive) and at least one lake-year observation in each decade from the 1970s through the 2010s (41 lakes, 1531 lake-years). We used the p-value for each lake-specific smooth term as a measure of whether the lake trend deviated significantly ($\alpha = 0.05$) from the shared, statewide trend.

Sensitivity of results to the time frame of analysis

To quantify the sensitivity of our results to the time frame of analysis, we used the statewide modeling approach to estimate trends in ice cover duration across time frames of varying lengths (10, 20, 30, 40, 50, 60, and 70 yr) and start dates (starting at 1949 and estimating a new trend with a new starting winter year for each year thereafter). We calculated the per-decade change between the first and last 5 yr of the time frame using the same methods described above.

Results

Model selection

Each ice metric was significantly related to winter year, winter year as an unordered factor, log-transformed lake area, lake ID, and UTM coordinates (Supporting Information Fig. S1; Table S1). No long-term pattern remained in the random annual intercept when the intercepts were ordered by winter year (Supporting Information Fig. S1g). Winter year as a random effect accounted for a large amount of variation in each ice metric (52% of variation in ice cover duration, 42% in ice formation, and 54% in ice breakup), reflecting synchronous annual ice cover dynamics among lakes. Notably, variance increased over time for each metric (Fig. 1a), which is reflected in model annual random intercepts (Fig. 1b) and, to a lesser extent, the significant heteroskedastic smooth function of standard deviation (Fig. 1c). Variance was also higher in southern Minnesota (Supporting Information Fig. S1d). Model residuals were randomly distributed over time (Fig. 1d).

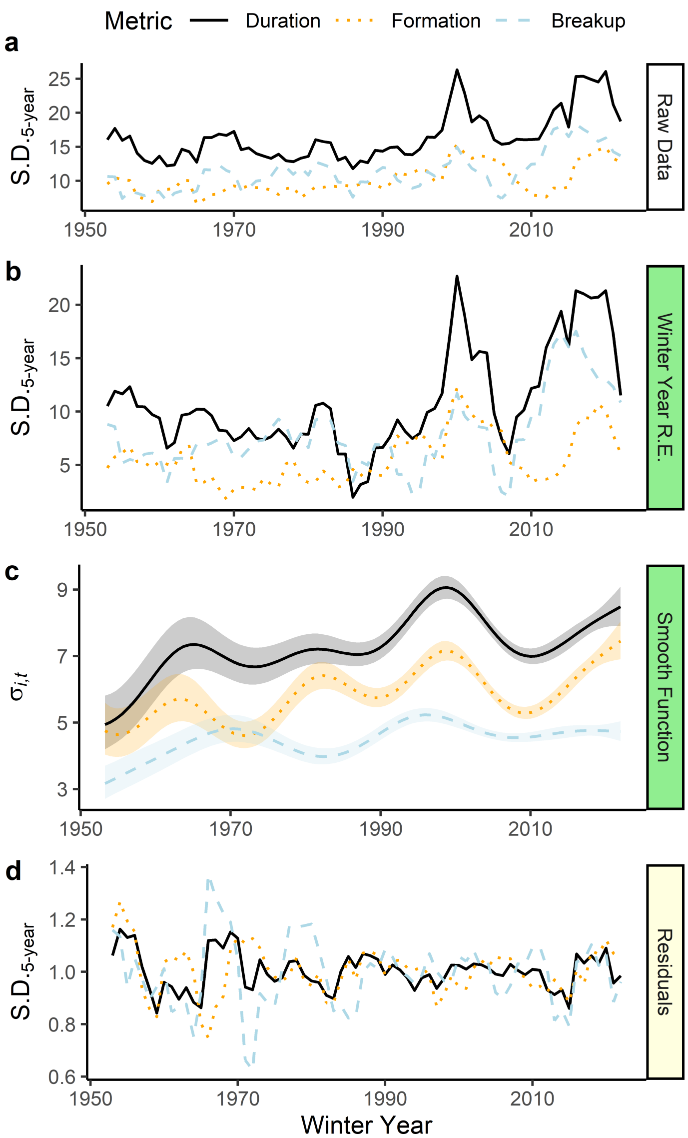

Fig. 1. Variance over time for ice cover duration, ice formation, and ice breakup as measured in (**a**) the 5-yr running standard deviation of the raw data, (**b**) the 5-yr running standard deviation of the annual random intercepts from the final models, (**c**) the temporal heteroskedastic smooth function from the final models (shaded areas represent 95% point-wise confidence intervals), and (**d**) the 5-yr running standard deviation of the model residuals.

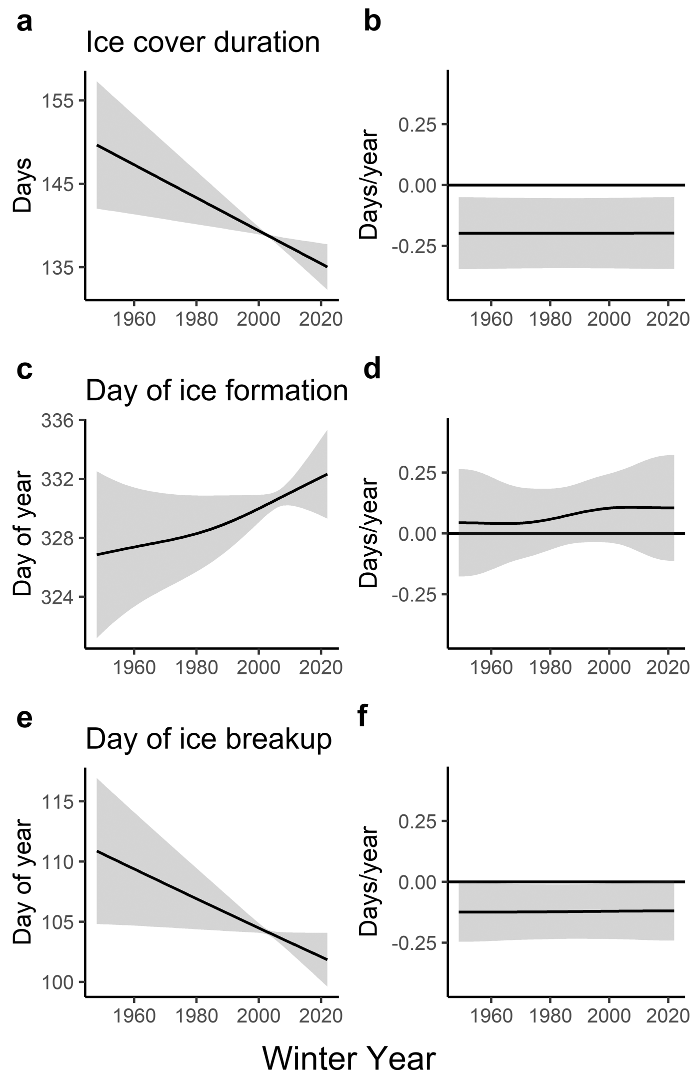

Fig. 2. Long-term trends (**a**, **c**, **e**) and derivatives (**b**, **d**, **f**) for ice cover duration (**a**, **b**), the day of ice formation (**c**, **d**), and the day of ice breakup (**e**, **f**) across all study lakes. Shaded areas represent 95% simultaneous intervals for the trend and derivative curves. Note that the *y*-axis in the derivative plots (**b**, **d**, **f**) can be multiplied by 10 to get the change in days/decade.

Statewide trends

The duration of ice cover in the average Minnesota lake declined at a constant rate of 2.0 d per decade from 1949 to 2022 (95% CI: 0.8–3.2; Fig. 2a,b; Table 1; estimated degrees of freedom of $f(\text{winter year}_t) = 1.0$). The day of ice formation shifted slightly later by 0.8 d per decade, though the CI included zero (95% CI: −0.4 to 1.9; Table 1). The long-term trend in ice formation date may be nonlinear with more positive change (advancing day) after the 1980s; however, the trend was uncertain (Fig. 2c,d) and the degrees of freedom of $f(\text{winter year}_t)$ was small (1.4). The day of ice breakup shifted earlier at a relatively constant rate (degrees of freedom of $f(\text{winter year}_t) = 1.02$) of 1.2 d per decade (95% CI: 0.5–1.9; Fig. 2e,f; Table 1). Trends and change estimates for ice duration, formation, and breakup using the whole dataset of 1213 lakes and 8778 lake-years were consistent when evaluated using a subset of data that included lakes with at least 10 (262 lakes; 5601 lake-years) and 30 yr (54 lakes; 2247 lake-years) of paired breakup and formation data, as well as for ice breakup using the larger unpaired dataset (1810 lakes; 18,636 lake-years) (Supporting Information Table S2; Fig. S2).

Lake-specific trends

Using the hierarchical model, we found that most of the 41 lakes (1531 lake-years) with at least 10 yr of observations and at least one observation in each decade from 1970 to 2022 followed roughly the same ($p > 0.05$) long-term trend as the shared statewide trend for each ice metric (85% for ice cover duration, 80% for day of ice formation, and 90% for day of ice breakup; Supporting Information Fig. S3) despite considerable among-lake variation in the predicted intercepts for each metric (Supporting Information Fig. S3).

Spatial patterns

Spatial patterns in each ice variable matched statewide patterns in air temperatures where ice cover duration was shorter, ice formation was later, and ice breakup was earlier in southern Minnesota lakes (Fig. 3a,c,e). However, we observed considerable local variation among lakes for ice cover duration and ice formation (Fig. 3). Lakes that are closer together possess less variation in ice breakup and cover duration than lakes that are further apart, resulting in strong spatial patterns (Fig. 3b,f), while variation between lakes increased only slightly with distance for ice formation (Fig. 3d).

Sensitivity of results to the time frame of analysis

We found considerable variation in the estimated ice cover trend and per-decade change estimates over time frames of analysis that were 40 yr or shorter; these estimates were often uncertain (Fig. 4). Notably, some trend estimates using recent time frames of up to 30 yr were positive, demonstrating increases in ice duration, and even 40-yr time frames led to differing estimates of ice loss depending on the time frame. Time frames of 50–70 yr produced variable trends, but estimates were centered around the 74-yr change estimate of −2.0 d per decade.

Discussion

Lake ice cover in Minnesota steadily declined by a total of 13.6 d from 1949 to 2022, a rate of 2.0 d per decade. This trend was nested within synchronous annual variability in each ice metric, and interannual variability in ice cover has increased over time. Despite strong spatial patterns and among-lake variation in each metric, we found that most lake-specific trends did not differ from the statewide trend. Finally, we found that trends in ice cover duration were sensitive to the time frame of analysis. This sensitivity and the caution it warrants are well established in the literature (Magnuson 1990; Johnson and Stefan 2006; Benson et al. 2012; Feiner et al. 2022), though to our knowledge, ours is the first study to explore this sensitivity continuously over a 70-yr time period. We quantify this sensitivity across a range of start and end dates and demonstrate that recent estimates of accelerating ice cover in Minnesota were likely due to shorter-term processes (e.g., from a period of rapid warming in the late 20th century; Jensen et al. 2007; Benson et al. 2012) and have not persisted through the early 21st century (Fig. 4). Instead of accelerating, ice cover loss in Minnesota lakes over the past 74 yr has been steady and consistent with estimates of well-documented global cases (1.7–1.9 d per decade; Woolway et al. 2020; Imrit and Sharma 2021; Sharma et al. 2021).

Table 1. Change in each metric was estimated using model simulations of early (1949–1953) and late (2018–2022) time periods and computing the difference between each 5-yr average. Confidence intervals (CIs) were calculated as the 2.5th and 97.5th quantiles of posterior simulations of the difference in 5-yr averages. Change over these two intervals was scaled to change in days per decade.

Metric	1949–1953	2018–2022	Change in days (95% CI)	Days per decade (95% CI)
Duration	149	135	−13.6 (−22, −5.3)	−2 (−3.2, −0.8)
Day of ice formation	327	332	5.1 (−2.5, 13)	0.8 (−0.4, 1.9)
Day of ice breakup	111	102	−8.4 (−13.3, −3.5)	−1.2 (−1.9, −0.5)

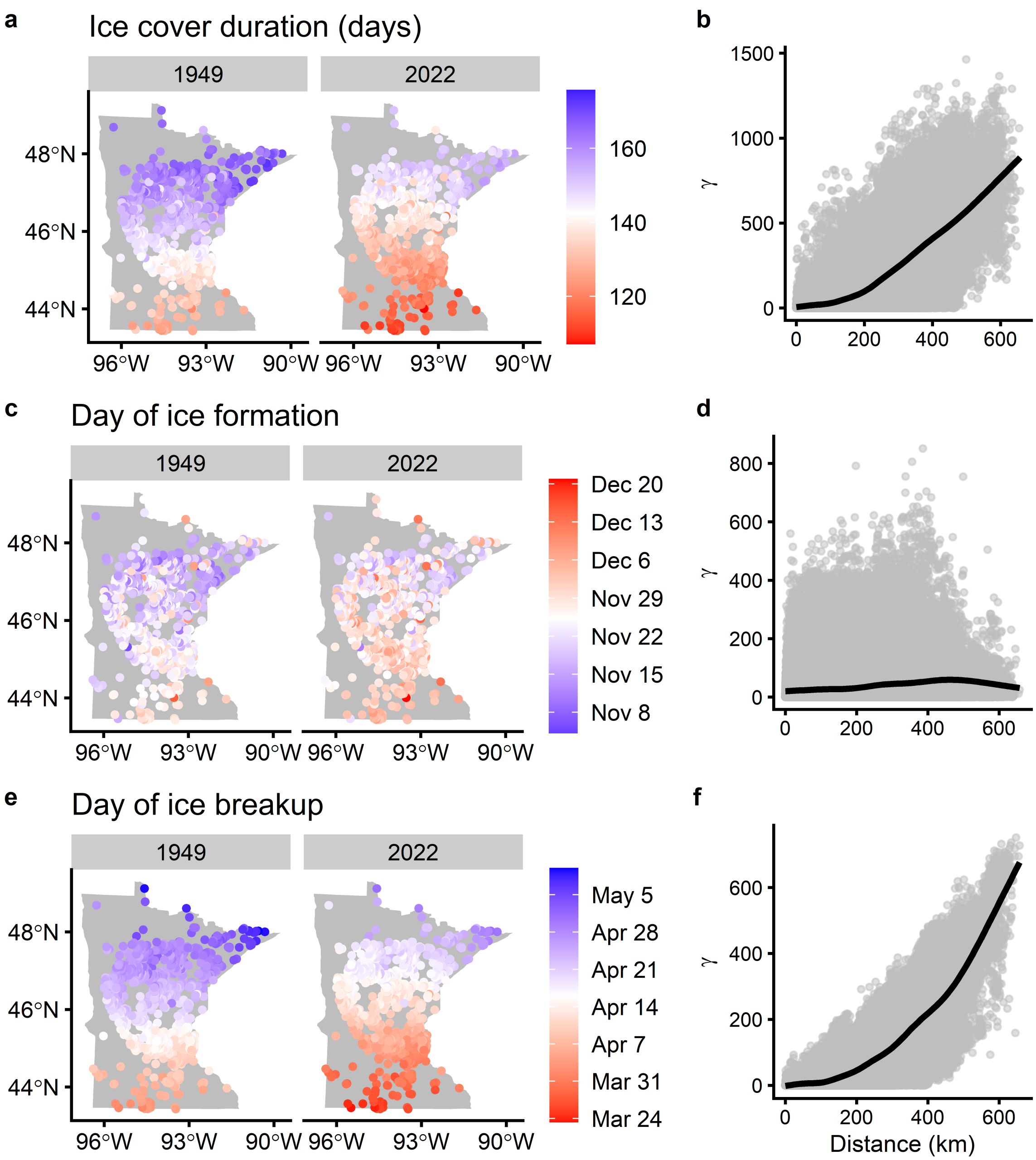

Fig. 3. Predictions for each model from early (1949) and late (2022) in the time series (**a**, **c**, **e**) and corresponding semivariograms (**b**, **d**, **f**) plotting the distance (in km) between two lakes and one half the variance of the difference between ice metric values for two lakes separated by that distance (γ). Red corresponds to shorter ice cover duration (**a**, blue = longer), a later day of ice formation (**c**, blue = earlier) and an earlier day of ice breakup (**e**, blue = later). White represents the median of the data, with each color representing minimum and maximum values.

We found little evidence to suggest that there are among-lake differences in long-term trends for ice metrics. In a study of ice cover dynamics of 678 northern hemisphere lakes and rivers, Newton and Mullan (2021) found that trends within a region were consistent with changes in regional average temperature. This high degree of synchrony is also reflected in our narrower study region. Additional research with a focus on the physical limnology of the 10–20% of lakes that differed from the statewide trend could shed light on why some lakes may respond differently to long-term change. Also, even with several dozen longer time series, the long-term trend estimates were uncertain (e.g., Fig. 2). As a result, lake-specific trends could be challenging to estimate and tease apart from the statewide trend, and we advise caution when comparing long-term trends among lakes to infer lake-specific responses to climate change.

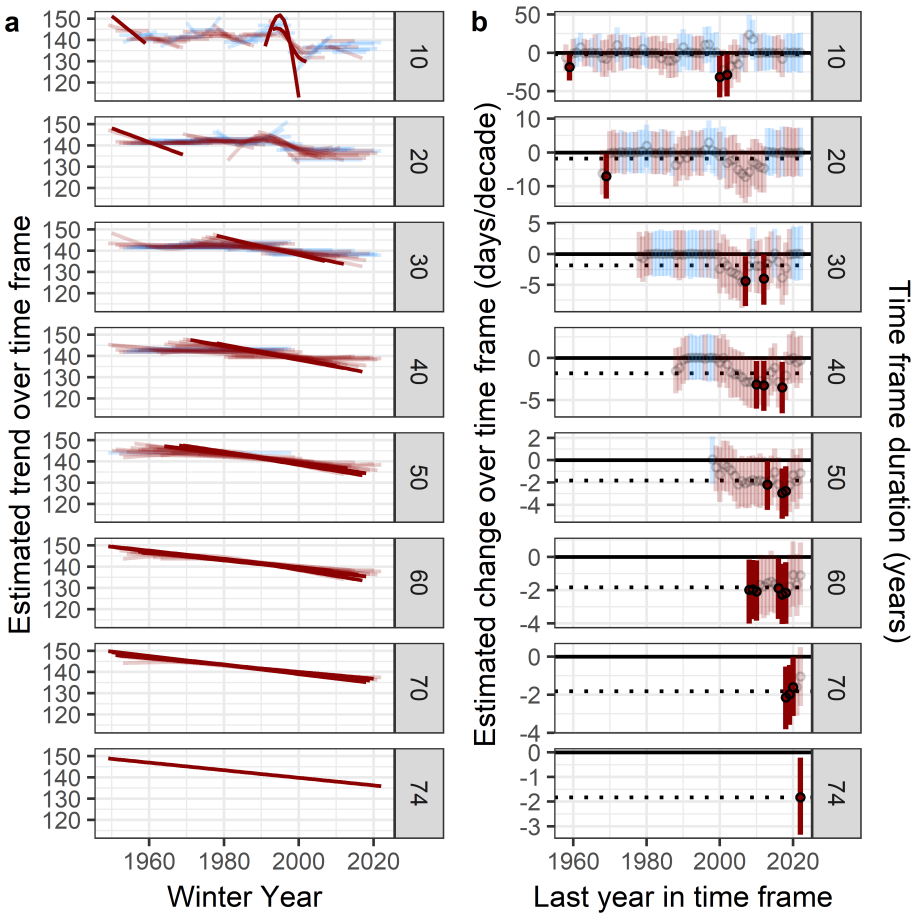

Fig. 4. Modeling approach here applied to time frames of varying durations (panel rows) and starting years created from the 1213 lakes and 8778 lake-years of Minnesota ice cover duration data. (**a**) Trend estimates over each time frame. (**b**) Estimated change in days per decade over each time frame. The dotted line is the −2.0 d per decade trend estimated using all ice cover duration data. Declining ice cover is in dark red, and increasing ice cover is in light blue. Opaque lines and filled circles highlight estimates with 95% confidence intervals (vertical lines in **b**) that do not include zero. Intervals include zero for estimates with transparent lines and open circles.

Rates of ice cover loss in the Northern Hemisphere are typically estimated using linear trend analysis, and such studies often report higher rates of ice cover loss in regressions estimated over or including recent time periods (e.g., 1996–2018; Woolway et al. 2020; Newton and Mullan 2021; Sharma et al. 2021) (Supporting Information Table S3). There have been similar periods of warming (and high variability) in Minnesota lakes, but accounting for random annual variation in our models suggested that these periods were not due to a permanent, accelerated response to warming. A similar analysis at the global scale could inform whether ice loss is accelerating in the Northern Hemisphere.

As linear regression has been the common approach for estimating trends over short and long periods, relatively few studies have investigated potential nonlinearity in trends as we did here. An exception is in the piecewise regression analysis of Imrit and Sharma (2021) where significant breakpoints revealed that ice cover loss was accelerating in five of 18 of their study lakes. Of the five with accelerating declines in ice cover duration, three (17% overall) were reported to have significant breakpoints in recent decades. These global estimates from piecewise regression are consistent with our results: ice cover loss in recent decades may be linear on average, and acceleration is not detected in most lakes.

In addition to contrasting with global observations of accelerating ice cover loss, our results of relatively steady ice cover loss from 1949 to 2022 in Minnesota contrast with reports of accelerating loss in recent decades in both Minnesota and the Upper Midwest (Weyhenmeyer et al. 2004; Johnson and Stefan 2006; Magee et al. 2016). Johnson and Stefan (2006) found accelerating lake ice loss in Minnesota due to ice formation and breakup shifting nearly twice as fast from 1990 to 2002 as the 1965–2002 study period (1979–2002 for ice formation). Our results, including more lakes and years, suggest that accelerating ice loss from 1990 to 2002 was likely due to short-term processes. Johnson and Stefan (2006) noted that trend reversals were common in their study and accelerating lake ice loss should not be extrapolated into the future. Indeed, trend reversals, accelerations, and decelerations over shorter time series durations were common here, even using our approach that accounts for random annual variation (Fig. 4). Apparent decelerations in recent decades have important implications for interpreting long-term change in datasets that begin in the 2000s or later, which would constitute most remotely sensed ice cover data (e.g., Zhang et al. 2021). Our results suggest ice cover dynamics will continue to be variable from year to year or even decade to decade (and possibly more over time with increasing variation in ice cover) and estimates of accelerating lake ice loss in recent decades may not be persistent.

Author Contributions

Gretchen J. A. Hansen conceived of the idea for the manuscript. Jake R. Walsh, Gretchen J. A. Hansen, and Kelsey Vitense designed the manuscript and provided leadership and oversight for the organization of the project and manuscript. Tarciso C. C. Leão developed an initial analytical framework to explore trends in ice duration over time. Jake R. Walsh, Christopher I. Rounds, Kelsey Vitense, Kenneth A. Blumenfeld, Holly K. Masui, and Andrew E. Honsey contributed to the development of the analytical framework used in the manuscript. Jake R. Walsh, Gretchen J. A. Hansen, Christopher I. Rounds, Kelsey Vitense, and Shyam M. Thomas contributed to paper development and writing. Jake R. Walsh, Christopher I. Rounds, Holly K. Masui, Shyam M. Thomas, Naomi S. Blinick, Jonah A. Bacon, and Claire L. Rude participated in literature review. Kenneth A. Blumenfeld and Peter J. Boulay contributed data. Jake R. Walsh, Christopher I. Rounds, Kelsey Vitense, and Holly K. Masui conducted data analysis, and Jake R. Walsh, Gretchen J. A. Hansen, Christopher I. Rounds, Kelsey Vitense, Kenneth A. Blumenfeld, and Peter J. Boulay helped interpret results. Jake R. Walsh, Gretchen J. A. Hansen, Christopher I. Rounds, and Kelsey Vitense drafted figures and tables, and Jake R. Walsh, Gretchen J. A. Hansen, and Christopher I. Rounds designed and created final versions of figures. Jake R. Walsh, Gretchen J. A. Hansen, Christopher I. Rounds, and Kelsey Vitense wrote manuscript text and edited the structure of the manuscript. All authors attended group meetings, participated in discussions, and performed critical review of the manuscript.

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

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